EXECUTIVE CLINICAL SCREENING SUMMARY

Original Retinal Fundus Photograph



PATIENT & SCREENING METADATA

- Patient / Examination ID: IDRiD$_0$01
- Examination Timestamp: 2026-10-14 22:24:36
- Screening System Engine: SHINRA Multi-Modal AI Engine V2
- Deployment Configuration: Rural Primary Health Center (Offline Mode)
- Image Resolution: 512 x 512 pixels
- Decision Governance: Calibrated Safety Gating Active (tau$_r$ef = 0.30)

1. IMAGE QUALITY SAFETY GATE

Quality Classification: GOOD

Composite Quality Score: 78.4 / 100

Safety Gradability Status: GRADABLE (Proceed with Autonomous Screening)

Operational Guidance: Image quality is clinically acceptable for automated AI screening.

2. DIABETIC RETINOPATHY DIAGNOSTIC FINDINGS

Predicted Severity: Grade 3 — Severe NPDR

Model Confidence: 78.3%

Expected Ordinal Score E[G]: 2.78 (Scale: 0.00 - 4.00)

Referable DR Decision: REFERABLE DR

Calibrated Referral Probability: 100.0% (Raw: 100.0% | Threshold tau$_r$ef = 0.30 | Concordance: 95.0%)

3. RELIABILITY SAFETY LAYER & CLINICAL TRIAGE DISPOSITION

RECOMMENDED ACTION: >>> ACCEPT SCREENING <<<

Operational Rationale: High reliability screening (90.0/100). Referable diabetic retinopathy predicted (P$_r$ef = 100.0%).

Composite Reliability Score: 90.0 / 100 (Tier: HIGH | Gate tau >= 60.0: PASSED)

Clinical Urgency: Specialist Ophthalmology Referral

Recommended Window: 2 to 4 weeks (Ophthalmology Clinic Referral)

4. DIAGNOSTIC SUMMARY & SAFETY LIMITATIONS

Diagnostic Summary: GOOD quality fundus photograph. System prediction indicates Grade 3 — Severe NPDR with 78.3% confidence and REFERABLE DR status (P$_c$al = 100.0%). Reliability is HIGH (90.0/100).

LIMITATIONS & DISCLAIMER: This screening report is generated by an artificial intelligence decision-support prototype designed for rural clinical triage. It does NOT constitute a definitive ophthalmological diagnosis. All automated findings must be verified by a qualified eye care professional before prescribing therapeutic interventions

CLINICAL EVIDENCE & BIOMARKER INVENTORY

RETINAL ANATOMICAL LANDMARKS

• Optic Nerve Head (Optic Disc):

- Geometric Center Location: (X = 381, Y = 322) pixels
- Disc Radius: 40 pixels
- Localization Confidence: 0.85 / 1.00 (Reliable Landmark)

• Macula & Foveal Risk Zone:

- Estimated Foveal Center: (X = 173, Y = 337) pixels
- Macular Threat Radius (1-DD): 120 pixels (2.5x Optic Disc Radius)
- DME Proximity Risk Factor: Grade 2 / 3 (Lipid exudates within macula)

RETINAL VASCULATURE ARCHITECTURE (DEEP U-NET)

• Vessel Segmentation Framework: Deep U-Net Trained Network Architecture

• Total Vessel Caliber Density: 2.26% of total retinal field

• Mean Vascular Tortuosity Index: 1.11 (Reference normal range: 1.05 - 1.25)

• Vascular Bifurcations / Branches: 7 detected vessel branch points

• Clinical Relevance in Diabetic Eye Disease:

- Vessel density attenuation indicates capillary drop-out and retinal non-perfusion.
- Elevated tortuosity index correlates with vascular remodeling and progression toward proliferative DR.

RETINAL LESION BIOMARKER INVENTORY

BIOMARKER	QUANTITATIVE MEASUREMENT	CLINICAL SIGNIFICANCE & STATUS
Microaneurysms (MAs)	101 lesions | 1023 px area	Earliest visible sign of NPDR; vascular pouching
Intraretinal Hemorrhages (HEs)	7 lesions | 3.91% area	Dot/blot capillary ruptures; severe NPDR marker
Hard Exudates (EXs)	58 lesions | 1.29% area	Lipid & lipoprotein extravasation from leaky capillaries
Macular DME Threat Grade	Level 2 / 3	Risk of Diabetic Macular Edema threatening central vision
Neovascularization Index (NV)	Biomarker Score: 0.32	Exploratory vessel morphometry (Not clinically certified)

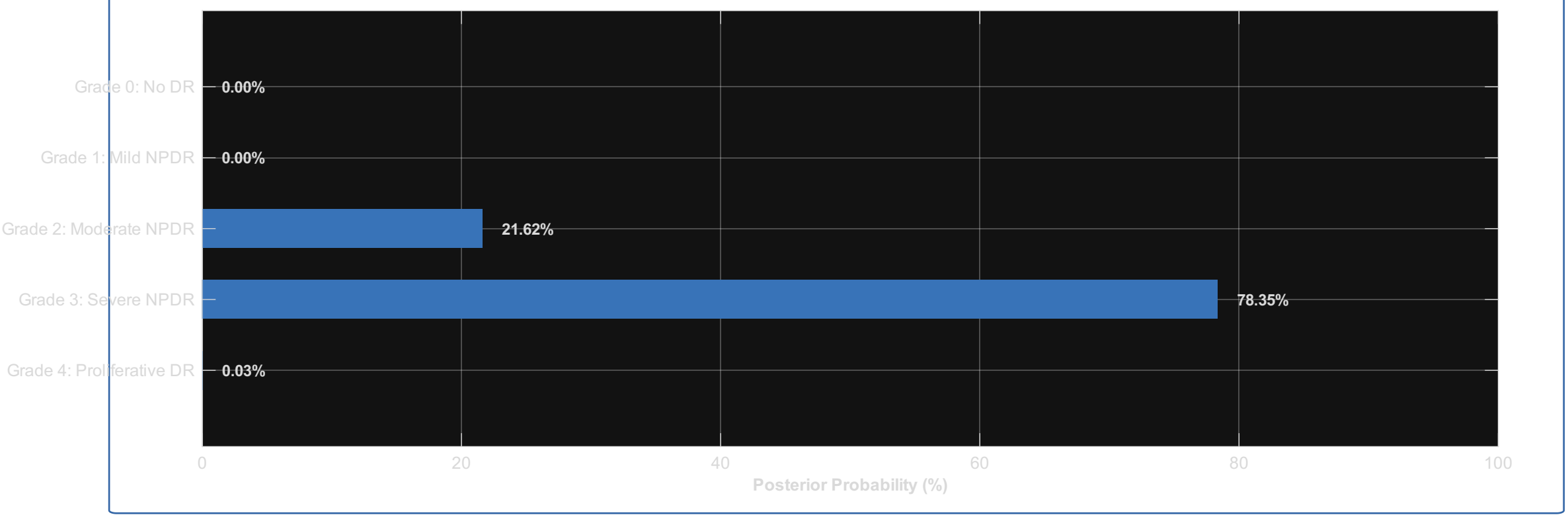
BIOMARKER EXTRACTION DISCLOSURE

• Microaneurysms, hemorrhages, and exudates are detected using calibrated multi-scale morphological filtering and CLAHE contrast enhancement.

• Neovascularization analysis is an exploratory biomarker derived from vascular branch complexity and does not replace fluorescein angiography.

MODEL OUTPUTS & 5-FACTOR RELIABILITY ANALYSIS

5-CLASS ICDR DR SEVERITY POSTERIOR PROBABILITY DISTRIBUTION



PROBABILITY CALIBRATION & THRESHOLD ANALYSIS

- Raw Referable DR Probability: 100.00% (Sum of Grade 2, 3, 4 posterior probabilities)
- Platt-Calibrated Probability: 100.00% (Fitted on 330-image development partition; a = 1.3106, b = -1.1412)
- Operating Referral Threshold: tau$_r$ef = 0.30 (Frozen development plateau operating point)
- Referral Screening Classification: REFERABLE DR (Calibrated P$_r$ef >= tau$_r$ef)
- Multimodal Concordance Metric: 95.0% agreement between CNN prediction and lesion burden

5-FACTOR ENGINEERING RELIABILITY BREAKDOWN

RELIABILITY COMPONENT	WEIGHT	SCORE [0-100]	FACTOR ASSESSMENT & RATIONALE
1. Image Quality (S_q)	25%	**78.4**	Classification: GOOD (Quality Score: 78.4)
2. Classifier Confidence (S_c)	25%	**74.0**	Max class posterior margin (Confidence: 78.3%)
3. Boundary Margin (S_m)	20%	**100.0**	Separation distance from referral threshold (|P - 0.30| = 0.700)
4. Biomarker Completeness ($S_c$omp)	15%	**100.0**	Integrity across quality, vessels, anatomy, lesions & CNN pipelines
5. Explainability Concordance ($S_a$gree)	15%	**97.5**	CAM activation and physical lesion overlap (Dice = 0.214)

COMPOSITE RELIABILITY: 90.0 / 100 | TIER: HIGH | SAFETY GATE: PASSED (tau$_a$ccept = 60.0)

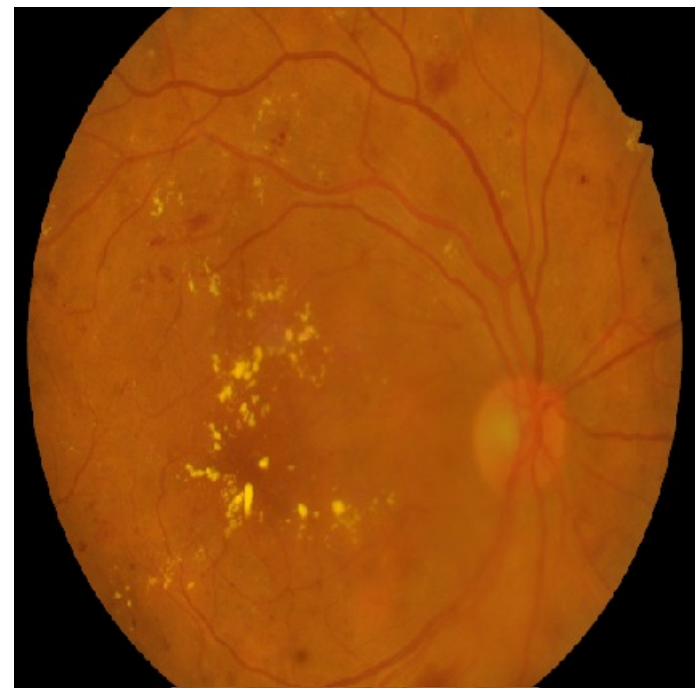
VISUAL INTERPRETABILITY ATLAS — PART 1



Original Fundus Photograph

True color 45-degree retinal fundus image as acquired from the camera.

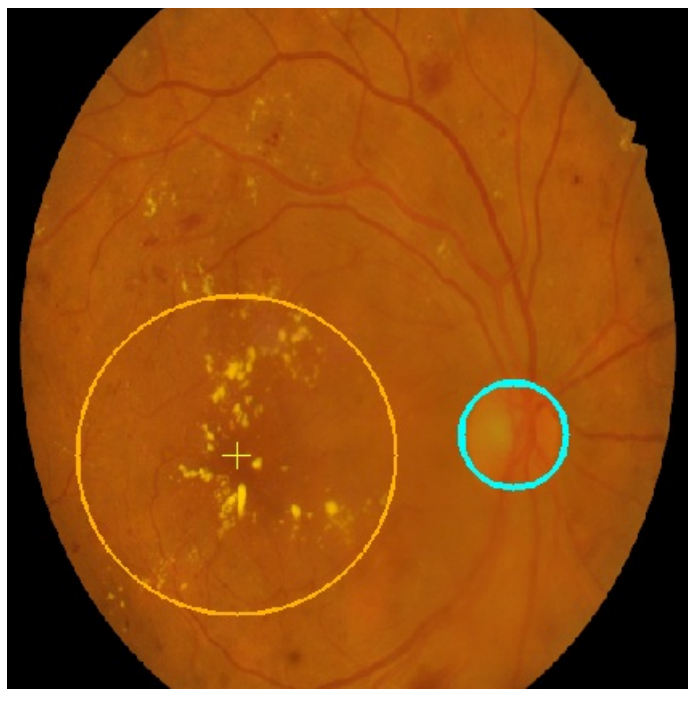
Legend: RGB Fundus Image



Preprocessed & Enhanced

Normalized illumination with shading correction and CLAHE contrast enhancement.

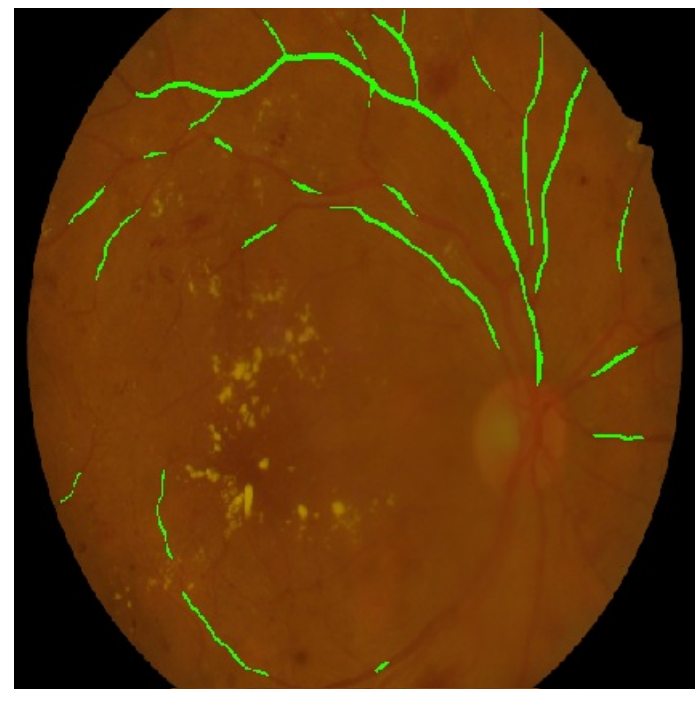
Legend: Contrast Enhanced



Retinal Anatomy

Cyan circle: Optic Disc boundary. Amber ring: Macular zone (2.5x OD radius).

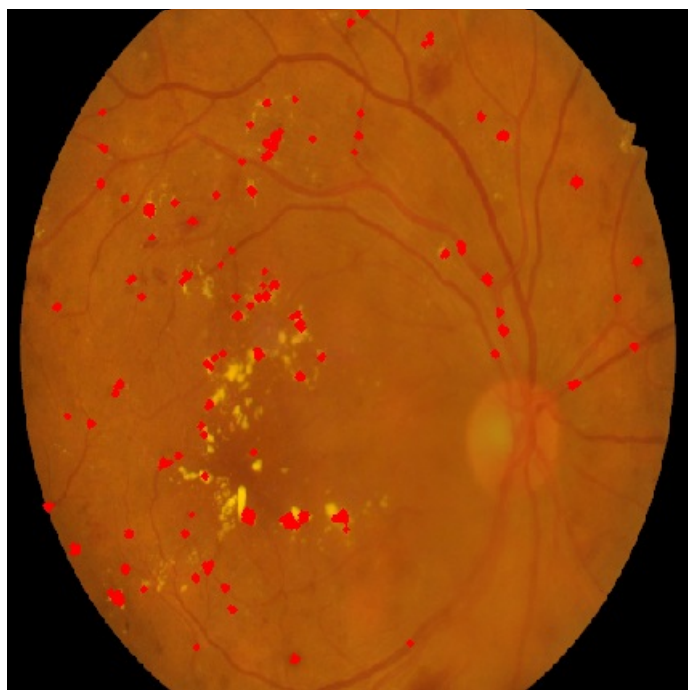
Legend: Cyan: OD | Amber: Macula



Retinal Vessel Architecture

Deep U-Net vessel network segmentation overlaid on normalized fundus.

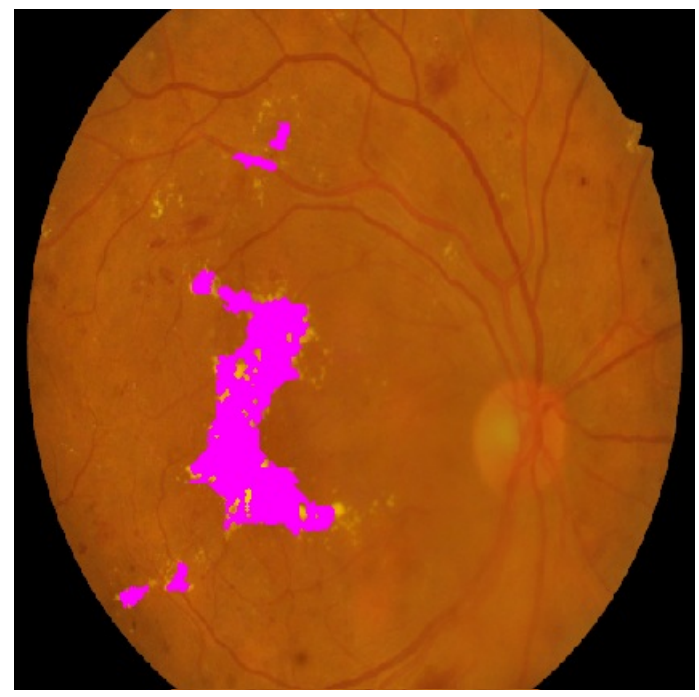
Legend: Green: Segmented Vessels



Microaneurysms Detection

Red bounding markers: Detected microaneurysm candidates.

Legend: Red: MAs (Morphology)

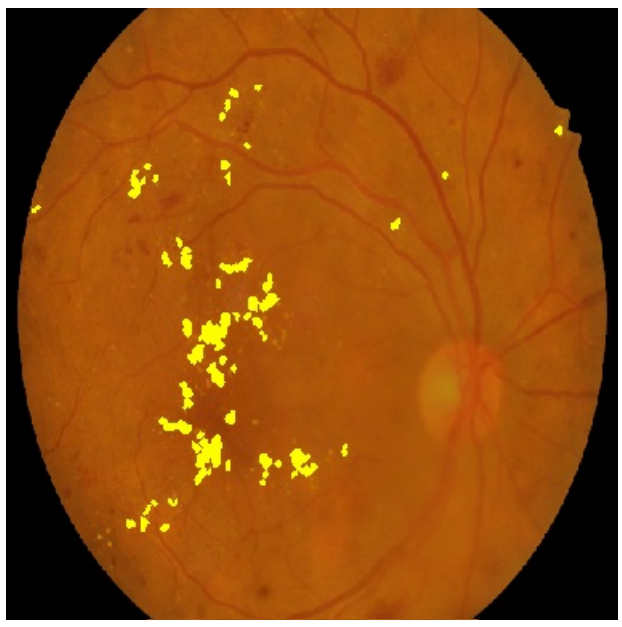


Hemorrhages Detection

Magenta regions: Detected intraretinal hemorrhages.

Legend: Magenta: Hemorrhages

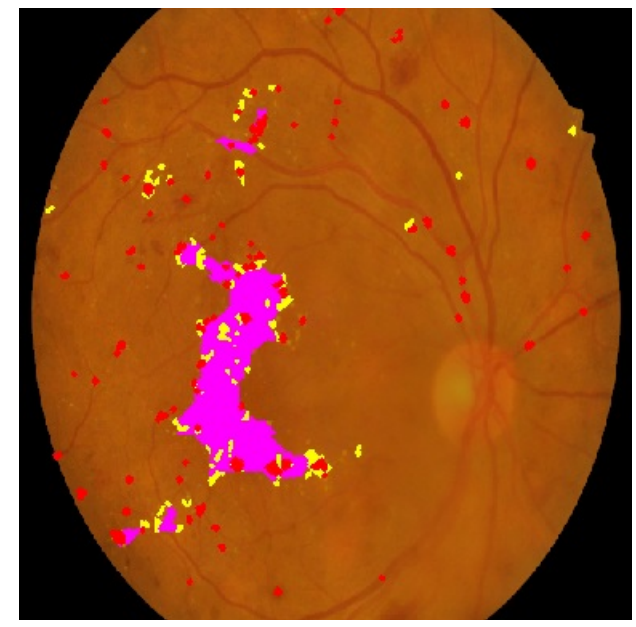
VISUAL INTERPRETABILITY ATLAS — PART 2



Hard Exudates & DME Risk

Yellow regions: Hard lipid exudates. Circles: Foveal proximity risk zones.

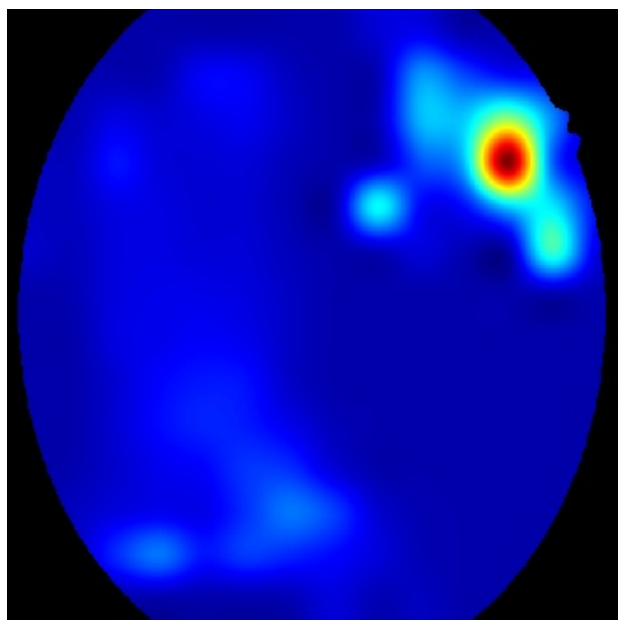
Legend: Yellow: Exudates | Blue: Fovea



Multi-Lesion Composite Map

Combined clinical biomarker overlay: Red (MA), Magenta (HE), Yellow (EX).

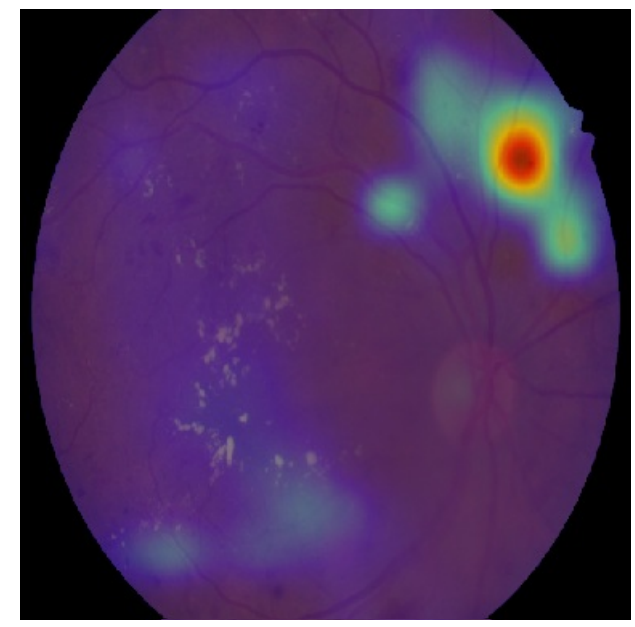
Legend: Red: MA | Magenta: HE | Yellow: EX



Grad-CAM Deep Heatmap

Class activation heatmap from SqueezeNet final convolutional layer.

Legend: Jet Colormap (Red=High attention)



Grad-CAM Clinical Blend

Alpha-blended Grad-CAM attention overlay providing spatial interpretability.

Legend: Alpha-Blend (0.5 Heatmap + Fundus)

CLINICAL TRIAGE & TELE-OPHTHALMOLOGY WORKFLOW SUMMARY

PRIMARY HEALTH CENTER PROTOCOL RECOMMENDATION:

- Examination Outcome: REFERABLE DR (Grade 3)
- System Triage Disposition: ACCEPT SCREENING
- Autonomous Decision Readiness: ACCEPTED (Meets high reliability threshold)
- Recommended Clinic Action: High-confidence referable DR detected. Schedule clinical examination with ophthalmologist.
- Recommended Referral Timeline: 2 to 4 weeks (Ophthalmology Clinic Referral)

DOCUMENTATION & AUDIT TRAIL:

- Software Engine Version: SHINRA Diagnostic Framework Release V2 (Development Frozen)
- Calibration Checkpoint: Platt Parameters a = 1.3106, b = -1.1412 (Fitted on IDRiD 330-train set)
- Operating Threshold: Referral tau_ref = 0.30 | Acceptance tau_accept = 60.0
- CAM-Lesion Concordance (Dice): 0.214 (Spatial alignment between neural attention and physical lesions)
- Report File Generated: d:/SHINRA/results/reports/test_report_IDRiD_01.pdf